STEPHEN LONG JR.

MECHANICAL ENGINEER | MECHANICAL CAD | FEA | CFD | ELECTROMECHANICAL DESIGN

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PROFESSIONAL SUMMARY

Mechanical Engineer with a multidisciplinary background in mechanical product design, CAD development, engineering simulation, thermal systems, and electromechanical integration. Experienced with 3D modeling and mechanical assemblies using SolidWorks, AutoCAD, CATIA, and Creo, alongside simulation workflows in ANSYS Fluent and ANSYS Mechanical for CFD, FEA, structural evaluation, and thermal analysis. Additional technical capabilities include PCB design and integration, ESP32-based embedded systems, Arduino IDE, C/C++ programming, sensor integration, and motor control. Interested in developing practical engineering solutions across mechanical products, robotics, energy systems, and connected electromechanical devices.

TECHNICAL SKILLS

Mechanical CAD & Product Design: SolidWorks, AutoCAD, CATIA, Creo, 3D modeling, parametric design, mechanical assemblies, engineering drawings

Engineering Simulation: ANSYS Fluent, ANSYS Mechanical, CFD, FEA, structural analysis, thermal analysis, heat transfer, fluid mechanics

Electronics & PCB: PCB design and layout, electronic circuit design, component integration, electronic enclosures

Embedded Systems & IoT: ESP32, Arduino IDE, C/C++, sensor integration, Bluetooth, Wi-Fi, motor control

Mechanical Engineering: Product development, prototyping, mechanical systems, robotics, renewable energy systems, HVAC, EV thermal management

Engineering Methods: Design review, technical documentation, simulation workflows, design refinement, system integration

PROFESSIONAL EXPERIENCE

Senior Mechanical Engineer

Singularis Solutions | Philadelphia, PA | March 2022 – August 2026

Technical Tools: SolidWorks, ANSYS Mechanical, ANSYS Fluent, AutoCAD, CFD, FEA

- **Mechanical Product Design & Development:** Developed mechanical product concepts and detailed 3D CAD models based on functional requirements, dimensional specifications, and manufacturing considerations. Worked on component geometry, material selection, mechanical interfaces, and system integration to support product development and design refinement.
- **Electromechanical Product Design:** Designed mechanical components and assemblies for electromechanical products, considering electronic interfaces, mounting arrangements, enclosure requirements, and functional integration. Evaluated component dimensions and mechanical compatibility to support product assembly and prototyping.
- **Gearbox Design & Structural Analysis:** Developed a detailed gearbox assembly incorporating spur gears, shafts, bearings, and housing components. Evaluated gear geometry, shaft alignment, mechanical clearances, and load transmission. Used ANSYS Mechanical to examine stress distribution and deformation under representative operating loads.
- **Mechanical Assembly Optimization:** Worked on mechanical assembly configurations and component positioning to improve design compatibility and simplify manufacturing and installation. Reviewed mating conditions, dimensional tolerances, interference issues, and component interfaces before refining CAD assemblies.
- **Prototype Development & Design Validation:** Supported prototype development through CAD modeling, component refinement, assembly preparation, and design evaluation. Reviewed mechanical interfaces, dimensional compatibility, and functional requirements to identify potential design improvements before further development.
- **EV Battery Cooling System Design & Thermal Analysis:** Designed a liquid-cooled battery thermal management concept incorporating battery modules, cooling plates, fluid channels, and inlet and outlet connections. Developed ANSYS Fluent simulation workflows to evaluate coolant flow behavior, pressure distribution, temperature variation, and heat transfer characteristics.
- **Mechanical Mechanism & Robotic Assembly Design:** Developed a six-axis robotic arm concept incorporating articulated links, mechanical joints, motor interfaces, transmission components, and an end-effector mounting arrangement. Created parametric CAD models and evaluated assembly integration, motion requirements, workspace limitations, and potential mechanical interference.

Mechanical Design & Simulation Engineer

Delserro Engineering Solutions | Easton, PA | June 2019 – February 2022

Technical Tools: SolidWorks, CATIA, ANSYS Fluent, ANSYS Mechanical, CFD, FEA, Thermal Analysis

- **Wind Turbine Design & CFD Analysis:** Developed a wind turbine assembly incorporating rotor blades, hub geometry, and supporting mechanical components. Evaluated aerodynamic design considerations and used ANSYS Fluent workflows to examine airflow patterns, velocity distribution, pressure variations, and wake behavior around the rotor.
- **HVAC Airflow Analysis:** Worked on mechanical design concepts for heating, ventilation, and air-conditioning systems, including ductwork, airflow passages, equipment interfaces, and supporting structures. Considered airflow distribution, thermal requirements, equipment arrangement, and integration with surrounding mechanical systems.

- **Industrial Fluid Flow System Development:** Developed fluid transport concepts involving piping arrangements, flow passages, connectors, and mechanical interfaces. Applied fluid mechanics principles and CFD analysis methods to evaluate velocity distribution, pressure variations, flow restrictions, and overall system behavior.
- **Solar Thermal Collector Design & Simulation:** Designed a solar thermal collector concept incorporating absorber components, collector tubes, manifolds, fluid passages, and structural supports. Evaluated thermal energy transfer and fluid circulation using thermodynamic principles and ANSYS Fluent simulation workflows.
- **Heat Exchanger Design & Thermal Analysis:** Developed heat exchanger concepts involving fluid channels, heat transfer surfaces, manifolds, and mechanical support structures. Evaluated geometry, flow passages, and thermal interactions to support effective heat transfer across industrial and energy-related applications.
- **Energy-Efficient Heat Pump Design:** Developed an air-source heat pump concept incorporating a compressor, evaporator, condenser, connecting piping, and equipment enclosure. Evaluated airflow and heat transfer around heat exchanger surfaces using thermal analysis and ANSYS Fluent workflows.
- **Thermal Management System Analysis:** Worked on thermal management concepts involving temperature regulation, heat dissipation, cooling arrangements, and fluid circulation. Evaluated thermal pathways, component positioning, temperature distribution, and operating conditions to support reliable thermal performance.
- **Structural Component Analysis:** Performed structural evaluation of mechanical components subjected to representative operating loads. Developed engineering models and used ANSYS Mechanical and FEA workflows to investigate stress concentrations, deformation, load distribution, and potential structural weaknesses.
- **Hydrogen Fuel Cell Stack CFD Analysis:** Developed a proton exchange membrane fuel cell stack concept incorporating bipolar plates, gas flow channels, end plates, and mechanical support components. Evaluated gas distribution, pressure variation, flow uniformity, and thermal behavior through CFD analysis workflows.

Mechanical & Electromechanical Engineer

Automation Research Group | Philadelphia, PA | August 2016 – May 2019

Technical Tools: SolidWorks, Creo, AutoCAD, PCB Design, ESP32, Arduino IDE, C/C++

- **Smart Autonomous Mini Robot Design:** Developed a compact autonomous mobile robot incorporating a mechanical chassis, motor mounts, wheels, sensor brackets, and electronic enclosure. Integrated ESP32-based control hardware with Arduino IDE and C/C++ programming for sensor processing, obstacle detection, motor control, and Bluetooth/Wi-Fi communication.
- **Industrial Automation Equipment Design:** Developed mechanical design concepts for automation equipment, including structural frames, component mounting arrangements, mechanical linkages, and equipment enclosures. Evaluated assembly requirements and mechanical interfaces to support integration of automated equipment.
- **Electronic Enclosure Design:** Designed mechanical enclosures for electronic assemblies while considering PCB dimensions, component clearances, connector positioning, ventilation, mounting requirements, and accessibility. Developed 3D enclosure models and evaluated mechanical compatibility with internal electronic components.
- **PCB Integration & Mechanical Packaging:** Worked on the integration of printed circuit boards into mechanical product assemblies. Evaluated PCB dimensions, mounting locations, connector access, component heights, enclosure clearances, and mechanical packaging requirements to support reliable electronic and mechanical integration.
- **Embedded Control System Development:** Developed embedded control concepts using ESP32 hardware, Arduino IDE, and C/C++ programming. Worked with sensor inputs, digital and analog interfaces, control logic, and communication functions to support interaction between electronic hardware and mechanical systems.
- **Sensor Integration & Motor Control:** Integrated sensors and motor control components into electromechanical prototypes. Worked with sensor signals, control logic, motor driver interfaces, and wireless communication methods, including Bluetooth and Wi-Fi, to support automated operation and remote monitoring.
- **Electromechanical Prototype Development:** Supported the development of electromechanical prototypes through mechanical modeling, electronic component integration, enclosure design, and assembly evaluation. Reviewed mechanical and electrical interfaces to identify compatibility issues and improve prototype integration.
- **Mechanical System Testing & Evaluation:** Supported mechanical and electromechanical system testing through design reviews, component compatibility checks, functional evaluations, and assembly inspections. Identified integration issues involving mechanical interfaces, electronic packaging, and operating requirements, and contributed to design refinement.

EDUCATION

Bachelor of Science in Mechanical Engineering

Drexel University | Philadelphia, PA, USA | September 2012 – May 2016

Field of Study: Mechanical Engineering